MonaSol Protocol

Cross-Chain Vault Architecture & Business Model

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Chapter 1 — Protocol Architecture

What MonaSol Is

MonaSol Protocol is a cross-chain vault custody system where assets are locked on Monad (EVM) and controlled exclusively via NFT keys on Solana. The user holds the NFT in their own Solana wallet at all times. MonaSol never holds the NFT, never holds a complete key, and has no custody of any unlock credential.

The core insight: MonaSol does not move money. It moves the deed to the vault. The NFT in the user's wallet plus a valid signature from that wallet are the only combination that opens a vault.

The Neighborhood Model

MonaSol uses a consistent analogy throughout all documentation, code, and user-facing language:

Analogy	Technical Term	Definition
Neighborhood	Protocol	The entire MonaSol system
Building	Locker	An isolated smart contract at its own on-chain address
Apartment	Vault	A user's asset container inside a Locker
Room	Sub-vault	A partitioned section inside a Vault with delegated access

Sub-vault is the technical term. Room is the user-facing name for the same concept.

Architecture — Eight Layers

1. Compartmentalized Risk

MonaSol deploys isolated Locker smart contracts — one per building in the neighborhood. Public Lockers hold up to 20,000 vaults. VIP Lockers hold as few as 10. Dedicated Lockers serve single institutions. No shared state between Lockers. A breach in one Locker has zero impact on any other.

2. Individual Vault Encryption

Every vault is independently secured by a cryptographic commitment tied to the corresponding Solana NFT. Compromising one vault reveals nothing about any other. Vault commitments use SHA-256 or SHA-512.

3. Sub-Vault Gating — The Rooms Model

Inside a single vault, the owner can create distinct Sub-vaults (Rooms), enabling granular delegated access without exposing the full principal. A CFO can be given access to Room A (payroll) while Room B (treasury reserve) remains mathematically inaccessible to their key.

4. Trustless Cross-Chain Verification — Solana Light Client on Monad

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MonaSol deploys a Solana Light Client directly on Monad. To unlock a vault, a user submits a Merkle proof of NFT ownership. The Monad contract verifies this cryptographically against the light client's known Solana state — no intermediary, no trusted third party, no admin key. This is novel infrastructure. Nobody has shipped a production Solana light client on an EVM chain. It is the longest item on the critical path.

5. On-Chain Opacity

Vault contents are not visible on the EVM side. An observer can see that a vault exists but cannot determine what is inside it without the owner's cryptographic Access Key. This is on-chain opacity — not zero-knowledge privacy. Pseudonymity is provided, not anonymity.

6. User-Owned Circuit Breakers

No platform-wide admin freeze. Every vault owner chooses System Mode (vault locks with other System-mode vaults in the same Locker when a threat is confirmed — pre-authorized collective protection) or Self Mode (user receives an alarm with Lock/Deny options; non-response triggers read-only degraded state). MonaSol can never unilaterally lock a Self-mode vault.

7. The Pledged State

Vault state machine: Active → Pledged → Settling → Released. To list an NFT for sale or use it as collateral, the vault must first enter Pledged state. This closes the front-running window — the seller cannot drain the vault while the transfer is in progress.

8. Node Health, Failover & Mandatory Rotation

2 Active Wallets (95% health threshold, 1 dedicated backup each) + 3 Approvers (90% threshold, shared pool of 2 backups). All backups dormant and encrypted until activated. Health score = (uptime + signature success) / 2, sampled every 10 minutes. Mandatory 120-hour rotation for all active nodes regardless of health. Collision guard prevents simultaneous rotation and failover on the same node.

Blast Radius Containment

Breach Level	Who Is Affected	Who Is Notified
Sub-vault	That sub-vault only	Vault owner — full alarm
Vault	That vault only	Vault owner — full alarm; same Locker owners — building watch
Locker	All vaults in that Locker	All Locker owners — full alarm; all other Lockers — neighborhood watch

Tech Stack

Layer	Technology	Role
Solana access layer	Rust & Anchor	NFT minting, state transitions, swap execution, circuit breakers
Monad storage layer	Vyper	Vault contracts, Locker factory, deterministic gas, formally verifiable
Cross-chain verification	Solana Light Client (Vyper on Monad)	Trustless NFT ownership proof — no oracle
Yield separation	Two-NFT model	Principal NFT (vault) + Yield NFT (interest stream) — isolated risk

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Layer	Technology	Role
NFT standard	Metaplex Core	~0.0029 SOL per mint, single-account design, 80% cheaper than legacy

Use Cases

1. Trustless Cross-Chain OTC Trading & Barter

Lock assets on Monad, mint Solana NFT key, pledge vault, list on Magic Eden. Buyer submits Merkle proof, vault transitions to Released. Atomic settlement — no bridge, no oracle.

2. Peer-to-Peer NFT Key Swap

Atomic swap — both NFTs transfer simultaneously or neither does. Flat fee. No vault contents displayed. No escrow. NexusBridge is never a counterparty.

3. Cross-Chain Institutional Custody

Institution locks assets in Dedicated Locker, holds NFT in Squads multisig. Capital changes hands by transferring the NFT — EVM assets never move.

4. Trustless Inheritance & Estate Planning

NFT placed in dead-man's-switch contract. Automatic transfer to heir on non-check-in. No lawyer, no court, no trusted executor.

5. Liquid Vesting

Unvested tokens locked in Lockers, NFTs airdropped to investors. Investors can sell the NFT (and future token rights) on secondary market without waiting for cliff.

6. DeFi Composability — Cross-Chain Collateral

Pledge vault, deposit NFT into SharkyFi. Borrow on Solana against Monad-locked collateral. No bridging, no slippage.

7. Event Ticketing — AI Screened & Identity Locked

Promoter deploys event Locker with one vault per tier. Tickets are numbered NFTs: #021*025-10 = seats 21–25, 5 admissions, $10 discount. Three anti-scalping layers: AI wallet screening, soul-bound KYC (Civic Pass), dual-token door check at venue.

8. Generational Trust & Yield Rights

Time-Capsule contract with unlock date. Cryptographic claim ticket — whoever holds it when the date arrives claims the NFT and vault. Separate Yield NFT can be leased to a third party while the principal remains locked.

Chapter 2 — Business Model, Token & Roadmap

Revenue Model

MonaSol earns revenue two ways only: a one-time lifetime lease at vault creation, and a flat fee at each transaction event. No fee references, percentages, or calculations touch the value or contents of any vault.

Lifetime Lease — Paid Once

Locker Tier	Vault Capacity	Lease (SOL)	USD @ $86
Public	Up to 20,000	0.05 SOL	~$4.30
Standard	Up to 1,000	0.15 SOL	~$12.90
VIP	10–100 vaults	0.50 SOL	~$43.00
Dedicated	Custom	Negotiated	Negotiated

The lease transfers with the NFT. When the NFT key is sold, the new owner inherits the permanent vault rights — no re-registration, no additional payment.

Per-Transaction Fee — Flat

Transaction	SOL	USD @ $86
Move-in	0.001 SOL	~$0.09
Deposit	0.001 SOL	~$0.09
Withdrawal	0.001 SOL	~$0.09
Swap	0.002 SOL	~$0.17
Transfer	0.001 SOL	~$0.09
Pledge / unpledge	0.001 SOL	~$0.09

Revenue at 50,000 Vaults — SOL Only (First 24 Months)

Tier	Vaults	Lease SOL each	SOL collected	USD @ $86
Public (70%)	35,000	0.05	1,750 SOL	$150,500
Standard (20%)	10,000	0.15	1,500 SOL	$129,000
VIP (8%)	4,000	0.50	2,000 SOL	$172,000
Dedicated (2%)	1,000	2.00	2,000 SOL	$172,000

Tier	Vaults	Lease SOL each	SOL collected	USD @ $86
Move-in fees	50,000	0.001	50 SOL	$4,300
TOTAL	50,000	—	7,300 SOL	$627,800

Lease and move-in revenue only — before a single deposit, withdrawal, swap, transfer, or event ticket is counted.

MSL Token

MSL is the native utility token of MonaSol Protocol with three functions: fee payment (from month 24), vault creation reward (locked 36 months from move-in date), and future staking for neighborhood watch participation. MSL is not a security, does not represent profit-sharing, and does not entitle holders to protocol revenue.

Total Supply: 2,000,000,000 MSL — Fixed Forever

Bucket	%	MSL	Purpose
Vault creation pool	50%	1,000,000,000	Distributed via vault creation, 36-month lock
Protocol treasury	20%	400,000,000	Development, operations, audits
Ecosystem & grants	15%	300,000,000	Builders on top of MonaSol
Liquidity	10%	200,000,000	DEX liquidity — Orca, Raydium
Team	5%	100,000,000	3-year vest, 1-year cliff

Vault Creation Reward — The Thirds Model

Every vault created receives MSL locked for exactly 36 months from that vault's creation date. The amount drops by one third each tier. Early adopters receive disproportionately more.

Tier	Vault Range	MSL/vault	Vaults	MSL Distributed	Cumulative MSL
1	1 – 500	36,000	500	18,000,000	18,000,000
2	501 – 1,500	24,000	1,000	24,000,000	42,000,000
3	1,501 – 3,500	16,000	2,000	32,000,000	74,000,000
4	3,501 – 7,500	10,667	4,000	42,668,000	116,668,000
5	7,501 – 15,500	7,111	8,000	56,888,000	173,556,000
6	15,501 – 31,500	4,741	16,000	75,856,000	249,412,000
7	31,501 – 63,500	3,160	32,000	101,120,000	350,532,000
8	63,501 – 127,500	2,107	64,000	134,848,000	485,380,000
9	127,501 – 255,500	1,404	128,000	179,712,000	665,092,000

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Tier	Vault Range	MSL/vault	Vaults	MSL Distributed	Cumulative MSL
10	255,501 – 511,500	936	256,000	239,616,000	904,708,000
11	511,501 – 767,500	624	256,000	95,292,000	1,000,000,000

Vault pool exhausts at approximately 767,500 vaults. After that vaults are created normally but receive no MSL allocation. Vault #001 receives 36,000 MSL. Vault #50,000 receives ~4,741 MSL. A 7.6x early adopter advantage permanently encoded in the protocol.

Payment Timeline

When	Event
Day 1	SOL payments only. NFT minting from Solana side only.
50,000 vault milestone	Cross-chain minting unlocks — users can move in from either Solana or Monad side.
Month 24	MSL payment activates for all services. 20% discount for MSL payers. SOL still accepted.
Month 36 (rolling)	First vaults hit 36-month mark. MSL unlocks on a rolling daily schedule — no cliff, no flood.
Month 36–54	Rolling unlocks complete across all early vaults. Pace mirrors original vault creation rate.

MSL Ecosystem

Staking — Weekly HODL Rewards

Users stake unlocked MSL post month 36. Weekly rewards from ecosystem bucket. Multipliers: flexible 1x, 3 months 1.5x, 6 months 2x, 12 months 3x, 24 months 5x. Staked MSL backs the neighborhood watch security layer — good behavior earns fees, malicious behavior is slashed.

MonaSol AI

AI inference layer where queries are paid in MSL. Native protocol context — vault history, tier, MSL balance. Use cases: vault configuration advice, plain-English smart contract summaries, event planning, neighborhood explorer queries. Every query burns MSL.

MonaSol Events

Protocol-run community events. Entry requires minimum MSL balance or staking position. Tickets are MSL-gated NFTs — MSL burned to claim (deflationary). Annual events: Founders Summit (vaults 1–500 only), Neighborhood Block Party (all active vault holders), Dedicated Locker dinner.

Merchandise

MSL-gated physical product drops. MSL burned per purchase — deflationary. Founder edition physical key ships free to vault #001–#500 holders. Limited runs tied to protocol milestones.

Designer NFT Collections

Collaborations with named artists — not AI generated. Limited collections purchasable exclusively in MSL. Holding one grants a permanent protocol benefit. Right artist collaboration crosses the crypto-art boundary into broader culture.

Partnerships

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MSL as payment layer for partner protocols. Priority: Magic Eden, Tensor, Monad DeFi protocols, hardware wallet manufacturers, legal and estate planning firms, event promoters.

Gaming — Virtual Neighborhood

Long-runway play. Virtual MonaSol neighborhood — buildings (Lockers), apartments (Vaults), rooms (Sub-vaults). MSL is in-game currency. Real vault assets have virtual representations. FiveM-style modular architecture — MonaSol provides the base world, community builds content.

How the Protocol Learns

Every promoter who deploys an event Locker teaches MonaSol what sells. Every artist who mints through the protocol teaches MonaSol what collectors value. Every institutional Dedicated Locker client teaches MonaSol what enterprise treasury managers actually need. Every gamer in the virtual neighborhood teaches MonaSol what the in-game economy requires.

This accumulated protocol intelligence — built from real usage — is the strongest moat MonaSol can build. By year 3, MonaSol knows things about cross-chain asset custody behavior that no research firm, no VC, and no competing protocol knows. The roadmap is a hypothesis. The community corrects it.

The Real Challenges

1. The Solana Light Client Doesn't Exist Yet

Nobody has shipped a production Solana light client running natively on an EVM chain. 6–12 months of specialist engineering minimum. Formal specification and independent security audit required before mainnet. Everything on the Monad side depends on it.

2. The Required Team Is Expensive and Rare

Vyper developers are rare. Anchor/Rust engineers who understand cross-chain verification are rarer. The gap between a working testnet and mainnet is 3–5 senior engineers at $150K–$300K per year each. The frontend is built. The engineering team is not.

3. Regulatory Exposure Is Real

The architecture is careful — on-chain opacity not zero-knowledge, flat fees not asset-based pricing, infrastructure not exchange operator. But soul-bound KYC for ticketing, verified transfer receipts, and institutional custody touch regulated territory across jurisdictions. Crypto-specialist legal review is required before taking institutional money.

4. 5-Party Consensus Is Operationally Complex

Active Wallets, Approver Ledgers, backup pools, 120-hour rotation, and health monitoring are live 24/7 infrastructure — not a smart contract you deploy and forget. DevOps infrastructure and on-call engineering are required from day one of mainnet.

5. 50,000 Vaults Is Not Guaranteed

50,000 vaults requires 50,000 real users making a deliberate decision to pay $5–$50 for a lifetime lease on a new protocol. Comparable protocols took 18–36 months. Some never reached it.

6. MSL Creates Regulatory Exposure

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A token that unlocks after 36 months and trades on DEXs walks a fine line with the SEC's Howey test. A legal opinion on MSL structure is required before public distribution. Launching outside the US first is the common approach.

The Funding Path

Stage	Source	Amount	Unlocks
Now	Solana Foundation grant	$50K–$100K	First Anchor engineer
Now	Monad ecosystem grant	$10K–$50K	First Vyper contract
Month 1–3	Hackathon prizes	$5K–$50K	Demo + team
Month 3–6	Superteam bounties	$1K–$10K	Community + visibility
Month 6–12	Pre-seed round	$500K–$2M	Full team, legal, light client
Month 12–24	Seed round	$2M–$5M	Mainnet, security audit, operations

Phased Roadmap

Phase	Month	Milestone	Funded by
0	Now	GitHub, Twitter, grant applications	Free
1	1–3	Monad testnet — basic Vyper Locker contract	Grant
1	1–3	Solana devnet — Anchor NFT minting program	Grant
2	3–6	Testnet demo — end-to-end move-in working	Grant
2	3–6	Pre-seed fundraising begins	Demo
3	6–12	Solana light client — formal specification	Pre-seed
3	6–12	Legal review — architecture and MSL token	Pre-seed
4	12–18	Light client — build and internal audit	Pre-seed
4	12–18	Security audit commissioned	Pre-seed
5	18–24	Mainnet launch — first public Lockers	Seed
5	18–24	50,000 vault milestone → cross-chain minting unlocks	Organic
6	24	MSL payment activates. Staking protocol launches.	Seed
7	24–30	Designer NFT collection. MonaSol AI beta.	Seed
8	36	First MSL unlocks rolling. Merch store opens.	Organic
9	36–48	Partnerships live	Organic

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Phase	Month	Milestone	Funded by
10	48–60	Gaming protocol alpha	Series A
11	60	Virtual neighborhood beta	Series A

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